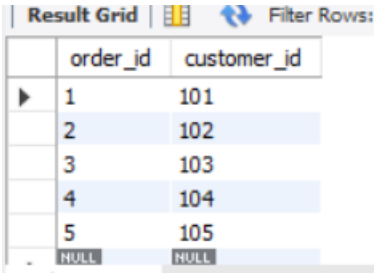
Week 5 – Payment Transaction Management System

1. Database & Payment Table Creation

A Payment Transaction Management System is used to store and manage payment details related to customer orders. The payment table stores payment ID, order ID, payment date, payment mode, payment status, and transaction amount. It helps maintain payment information in an organized manner.



Result Grid | Filter Rows:

order_id	customer_id
1	101
2	102
3	103
4	104
5	105
NULL	NULL

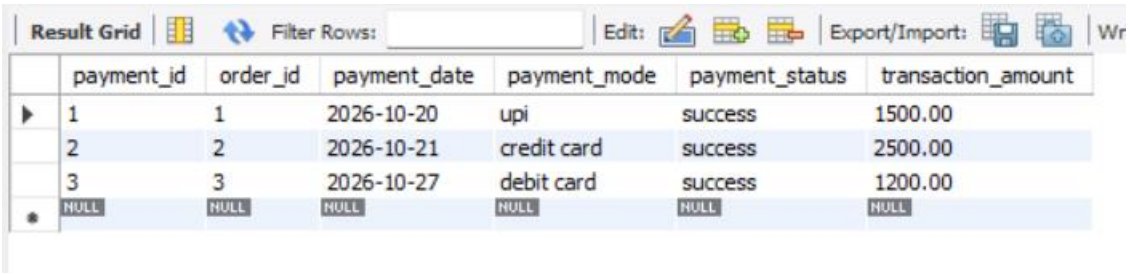
2. Primary Key, Foreign Key & Other Constraints

Constraints are rules used to maintain the accuracy and integrity of data in a database.

- **Primary Key:** Uniquely identifies each payment record.
- **Foreign Key:** Connects the payment table with the orders table.
- **Not Null:** Ensures that required fields cannot be empty.
- **Unique:** Prevents duplicate payment records for the same order.
- **Check:** Ensures that only valid payment modes, statuses, and transaction amounts are accepted.

3. Sample Data – INSERT Queries

- Sample data is used to test the payment management system. Payment records contain details such as order ID, payment date, payment mode, payment status, and transaction amount. Different payment modes include UPI, Credit Card, Debit Card, Net Banking, and Cash on Delivery.



Result Grid | Filter Rows: | Edit: | Export/Import: | Wr

payment_id	order_id	payment_date	payment_mode	payment_status	transaction_amount
1	1	2026-10-20	upi	success	1500.00
2	2	2026-10-21	credit card	success	2500.00
3	3	2026-10-27	debit card	success	1200.00
NULL	NULL	NULL	NULL	NULL	NULL

4. Payment Transaction CRUD Queries

CRUD stands for Create, Read, Update, and Delete. These operations are used to manage payment records in the database.

- **Create:** Adds new payment transactions.
- **Read:** Displays and retrieves payment details.
- **Update:** Modifies existing payment information.
- **Delete:** Removes payment records when necessary.

Result Grid | Filter Rows: | Edit: | Export/Import:

payment_id	order_id	payment_date	payment_mode	payment_status	transaction_amount
1	1	2026-10-20	upi	success	1500.00
2	2	2026-10-21	credit card	success	2500.00
3	3	2026-10-27	debit card	success	1200.00
4	4	2026-10-23	net banking	pending	3000.00
5	5	2026-10-24	cash on delivery	pending	800.00

5. Transaction Status Management Queries

Transaction status management is used to monitor the current state of a payment. The payment status can be Pending, Success, or Failed. It helps identify completed payments, unsuccessful transactions, and payments that are awaiting confirmation. The status can be updated after a transaction retry.

Result Grid | Filter Rows:

payment_mode	total_revenue
upi	1500.00
credit card	2500.00
debit card	1200.00

6. Payment Analysis & Revenue Report Queries

Payment analysis is used to study transaction details and generate reports. It helps calculate the number of transactions by payment mode, identify the most preferred payment method, calculate successful payment revenue, find the average transaction amount, and view customer payment history. It also helps count successful, failed, and pending transactions and identify orders with payment issues.

Result

The Payment Transaction Management System helps store, manage, and analyze payment records efficiently. It maintains the relationship between orders and payments and provides useful transaction and revenue reports.